COMP108
Data Structures and Algorithms

Dynamic Programming (Part II Assembly Line Scheduling)

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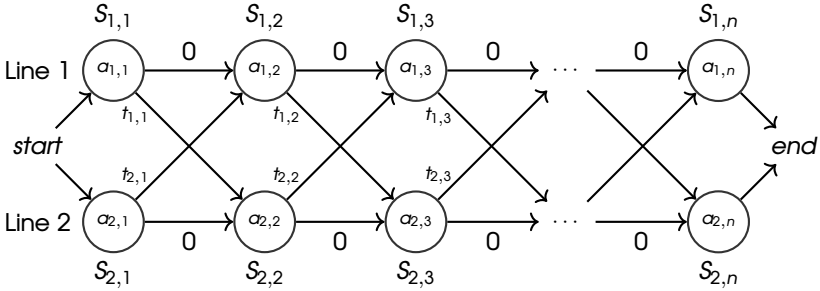
2025-26

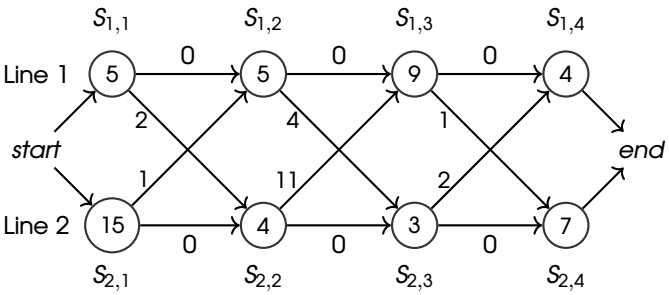
Assembly Line Scheduling

- 2 assembly lines, each with n stations ($S_{i,j}$: line i station j)
- $S_{1,j}$ and $S_{2,j}$ perform same task but time taken is different
- $a_{i,j}$: assembly time at $S_{i,j}$
- $t_{i,j}$: transfer time from $S_{i,j}$

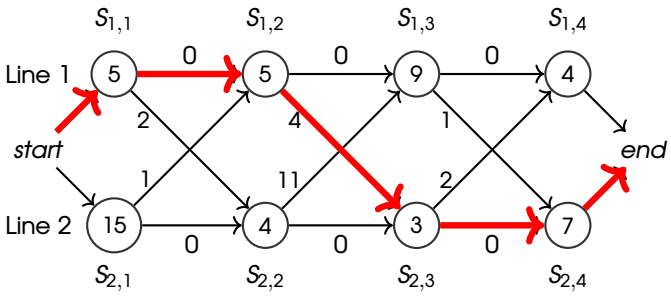


Problem: To determine which stations to go in order to **minimize** the total time through the n stations

Example

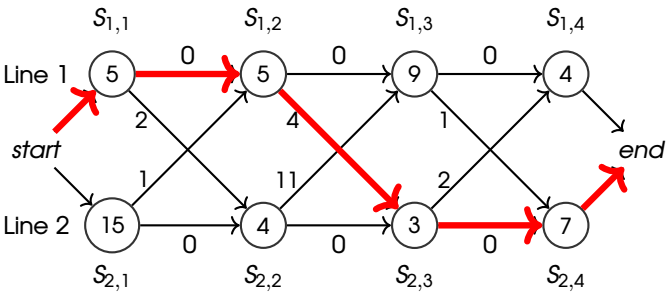

$S_{1,1}$
$S_{1,2}$
$S_{1,3}$
$S_{1,4}$
Line 1
Line 2
start
end
5
5
9
4
15
4
3
7
0
0
0
2
4
1
1
11
2
0
0
0
$S_{2,1}$
$S_{2,2}$
$S_{2,3}$
$S_{2,4}$

Example



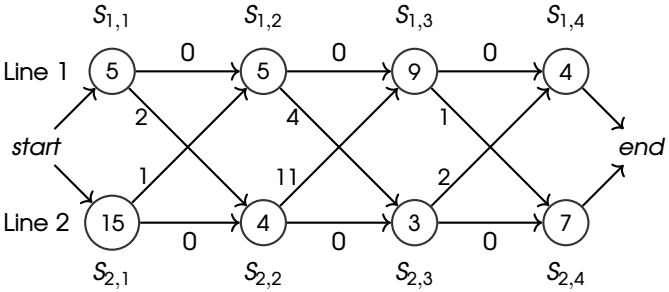
station chosen	$S_{1,1}$	$S_{1,2}$	$S_{2,3}$	$S_{2,4}$

Example



station chosen	$S_{1,1}$	$S_{1,2}$		$S_{2,3}$	$S_{2,4}$	
time required	5	5	4	3	7	= 24

Example


$S_{1,1}$
$S_{1,2}$
$S_{1,3}$
$S_{1,4}$
Line 1
5
5
9
4
0
0
0
2
4
1
start
end
1
11
2
Line 2
15
4
3
7
0
0
0
$S_{2,1}$
$S_{2,2}$
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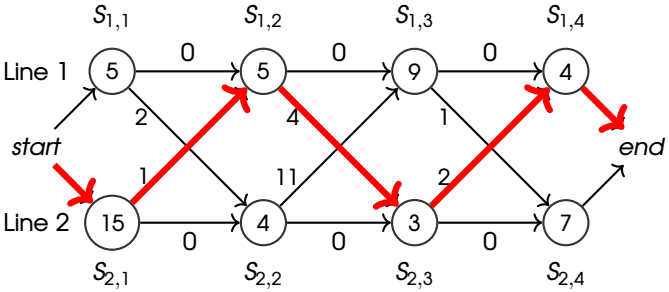
Example



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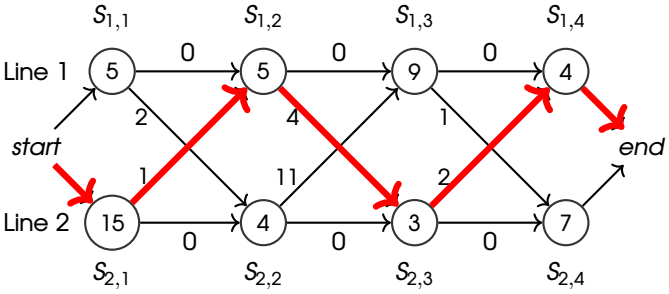
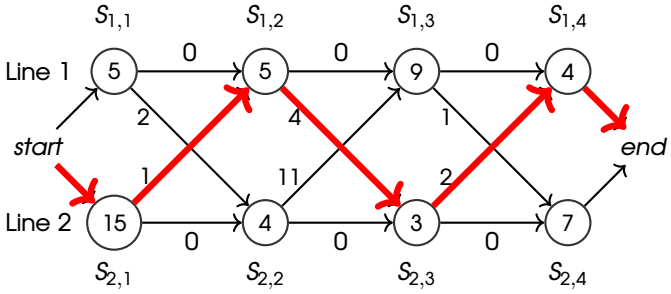


station chosen	$S_{2,1}$		$S_{1,2}$		$S_{2,3}$		$S_{1,4}$	
time required	15	1	5	4	3	2	4	= 34

Example



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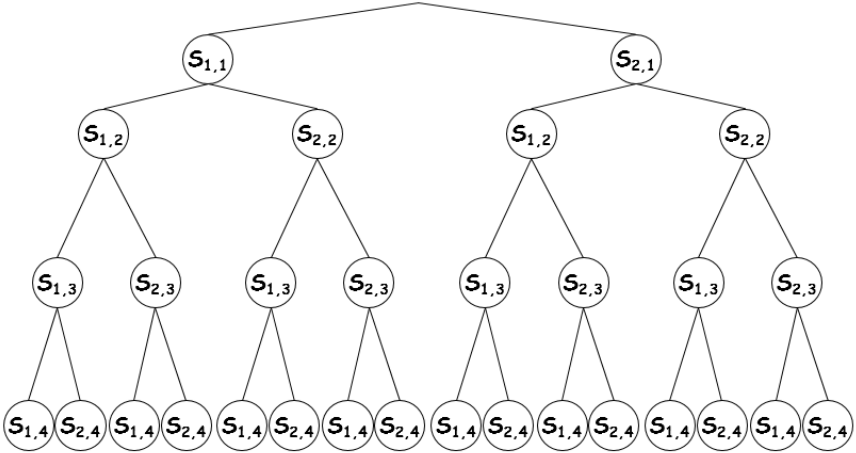
How to determine the best stations to go?

There are altogether 2^n choices of stations.

Should we try them all?

All possible choices



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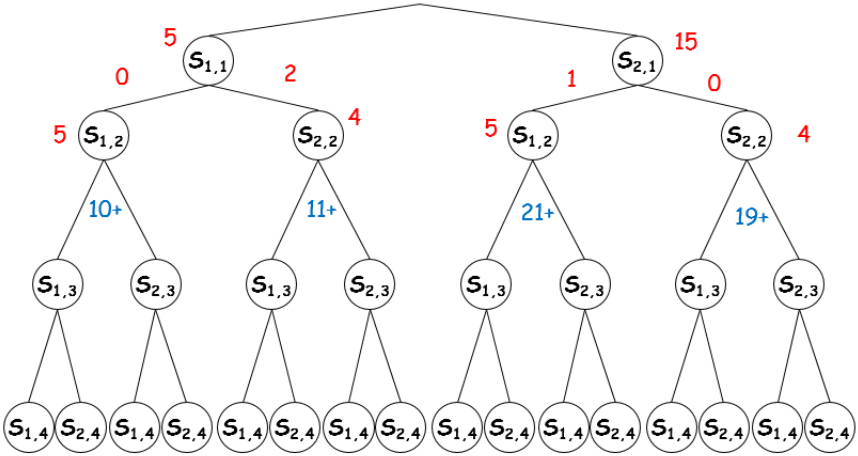

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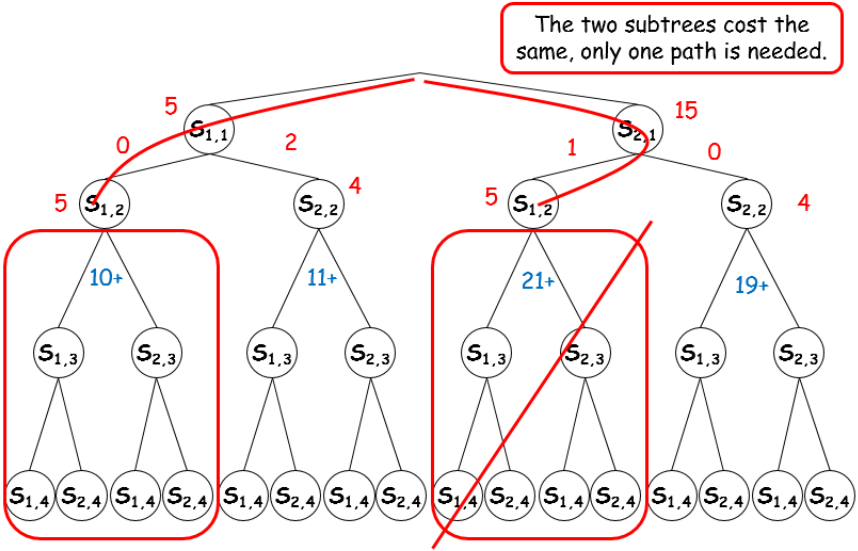

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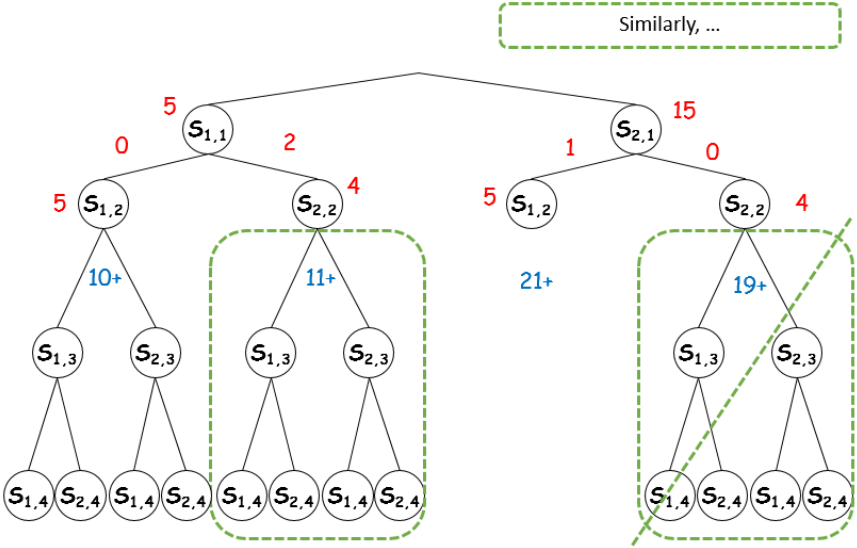

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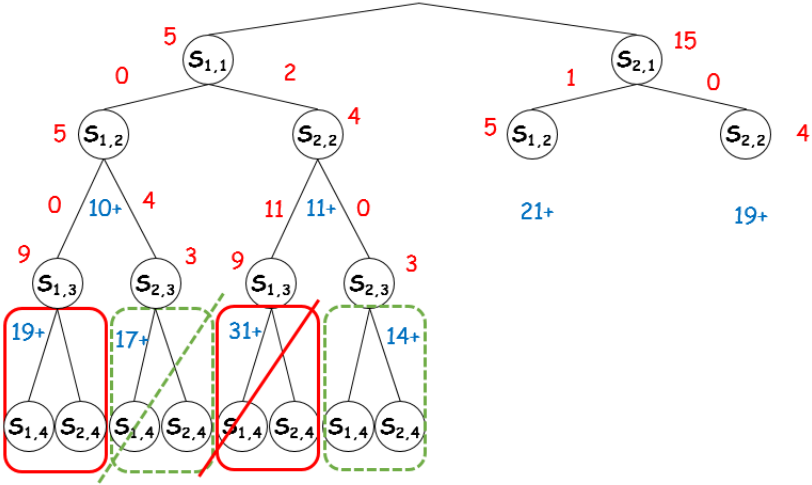

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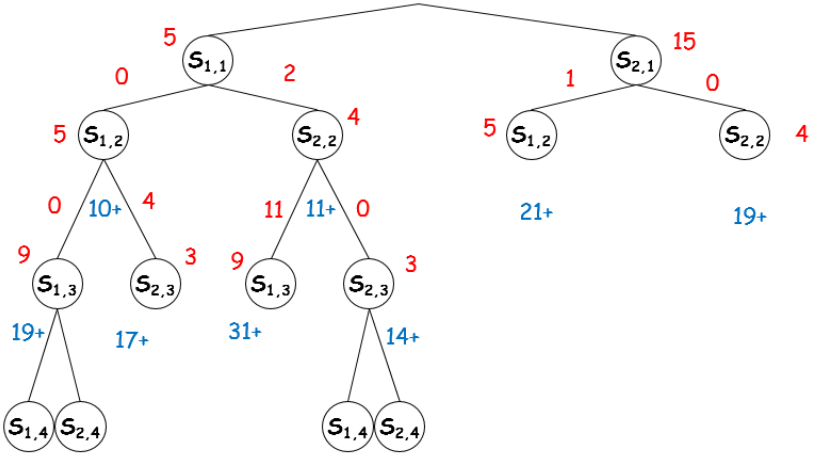

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  1. If we know the fastest ways to get thro' station $S_{1,n}$ and $S_{2,n}$
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  3. Similarly for $S_{2,n}$
  4. Generalsing, we want fastest way to get thro' $S_{1,j}$ and $S_{2,j}$, for all j.

A dynamic programming solution - formalisation

What are the sub-problems?

- given j, what is the fastest way to get thro' $S_{1,j}$
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Task:

- Starting from $f_1[1]$ and $f_2[1]$, compute $f_1[j]$ and $f_2[j]$ incrementally
- i.e., $f_1[2]\&f_2[2]$, $f_1[3]\&f_2[3]$, $\cdots$, $f_1[n]\&f_2[n]$

Solving the sub-problems (1)

Q1: What is the fastest time to get thro' $S_{1,j}$

A: either

- the fastest way thro' $S_{1,j-1}$, then directly to $S_{1,j}$, or
- the fastest way thro' $S_{2,j-1}$, a transfer from line 2 to line 1, and then thro' $S_{1,j}$

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(i) $f_1[j-1] + a_{1,j}$

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(i) $f_1[j-1] + a_{1,j}$ (ii) $f_2[j-1] + t_{2,j-1} + a_{1,j}$

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$$\text{(i) } f_1[j-1] + a_{1,j} \qquad \text{(ii) } f_2[j-1] + t_{2,j-1} + a_{1,j}$$

$$\therefore f_1[j] = \min\{f_1[j-1] + a_{1,j}, \quad f_2[j-1] + t_{2,j-1} + a_{1,j}\}$$

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$$\text{Boundary case: } f_1[1] = a_{1,1}$$

Solving the sub-problems (2)

Q1: What is the fastest time to get thro' $S_{2,j}$

A: either

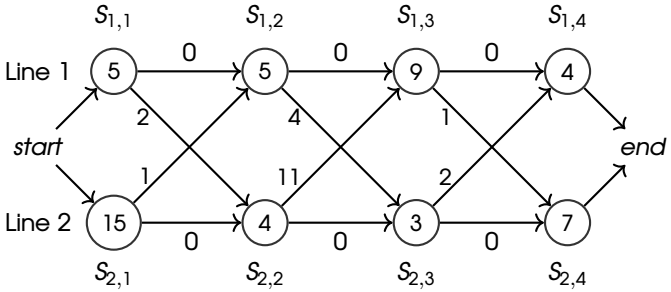
- the fastest way thro' $S_{1,j-1}$, a transfer from line 2 to line 1, and then thro' $S_{2,j}$, or
- the fastest way thro' $S_{2,j-1}$, then directly to $S_{2,j}$

$$\text{(i) } f_1[j-1] + t_{1,j-1} + a_{2,j} \qquad \text{(ii) } f_2[j-1] + a_{2,j}$$

$$\therefore f_2[j] = \min\{f_1[j-1] + t_{1,j-1} + a_{2,j}, \quad f_2[j-1] + a_{2,j}\}$$

$$\text{Boundary case: } f_2[1] = a_{2,1}$$

Example again

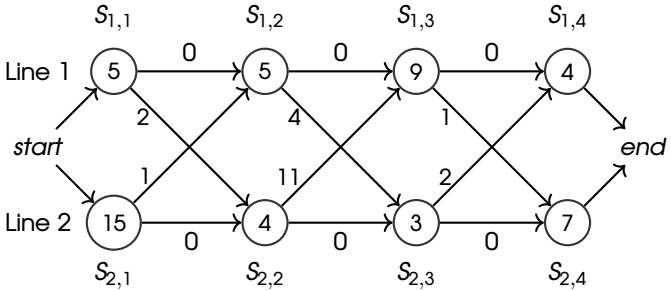


$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$
5			

minimum cost: _____

$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$
15			

Example again

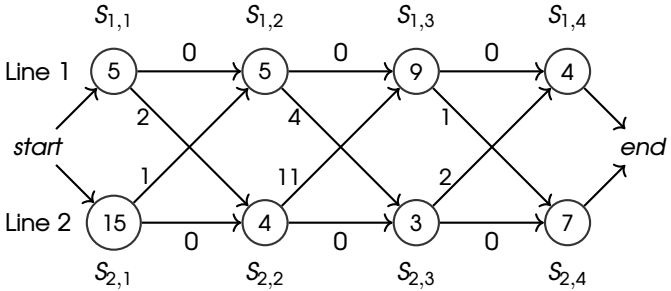


$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$
5 $\longrightarrow$	5+5=10		

minimum cost: _____

$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$
15			

Example again

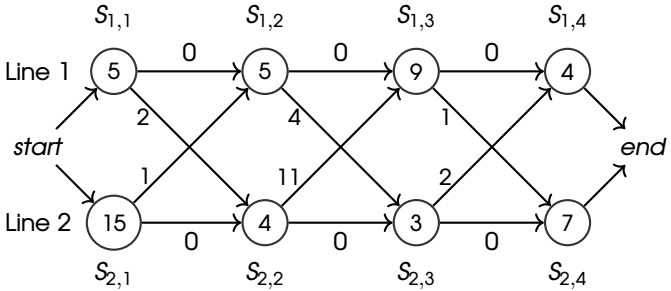


$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$
5	5+5=10		
	15+1+5		

minimum cost:

$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$
15			

Example again

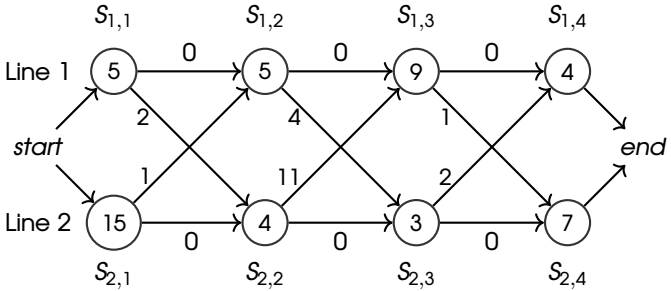


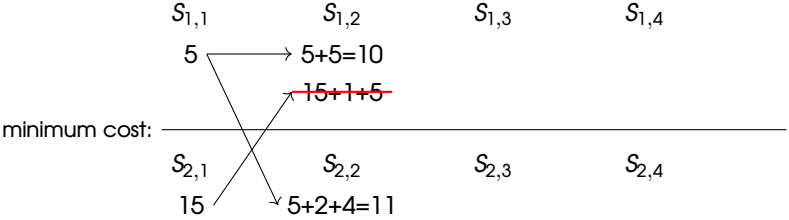
$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$
5	5+5=10		
	~~15+1+5~~		

minimum cost:

$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$
15			

Example again





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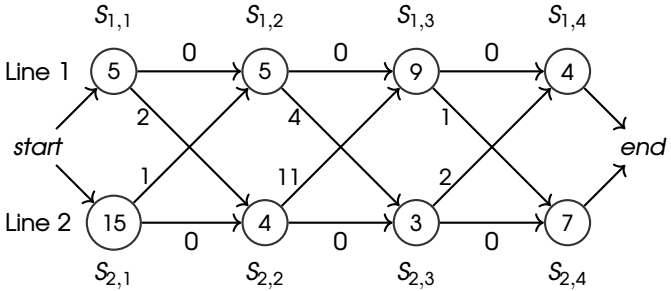
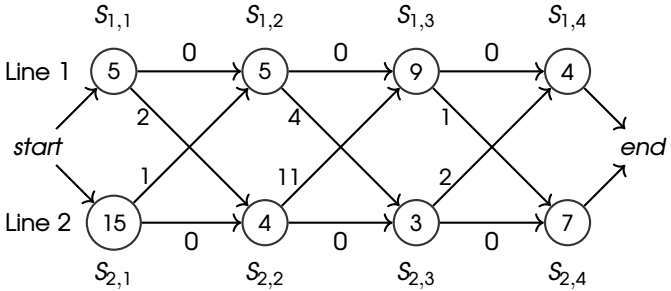


$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$
5	5+5=10		
	~~15+1+5~~		
minimum cost:			
$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$
15	5+2+4=11		
	15+4		

Example again



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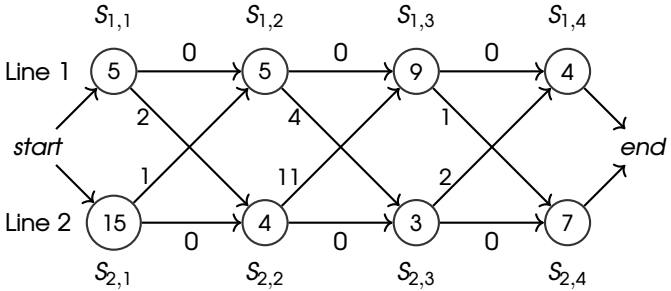
Example again



$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$
5	5+5=10	10+9=19	
	~~15+1+5~~		

minimum cost:

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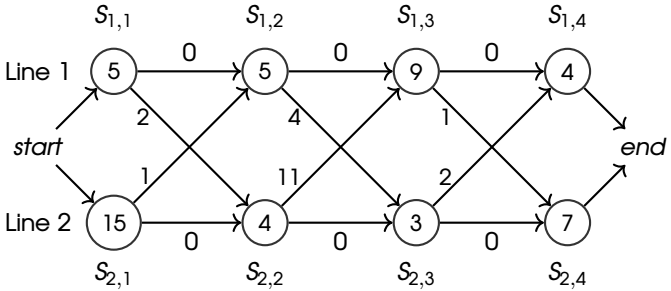
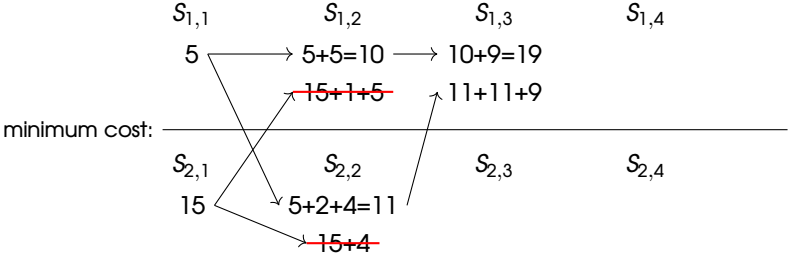
15	5+2+4=11		
	~~15+4~~		

Example again





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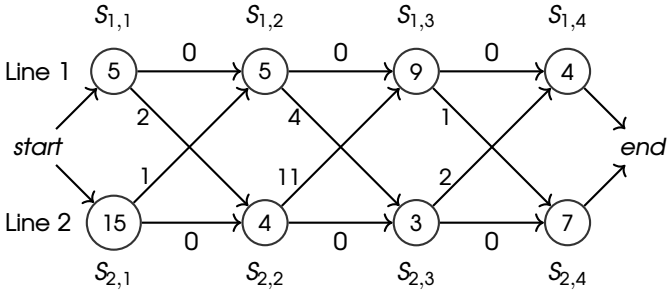

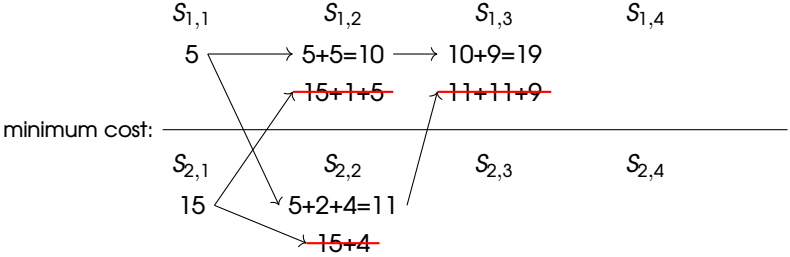

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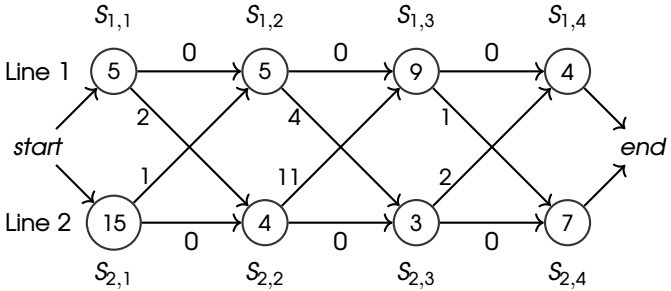

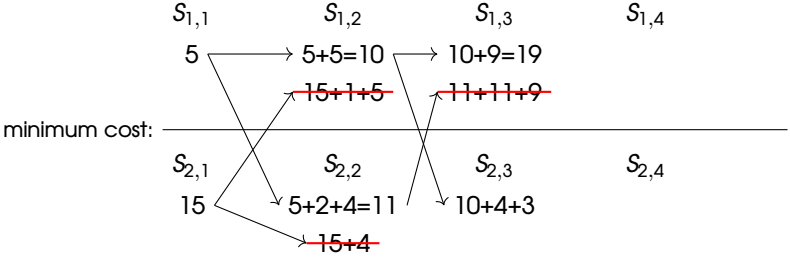

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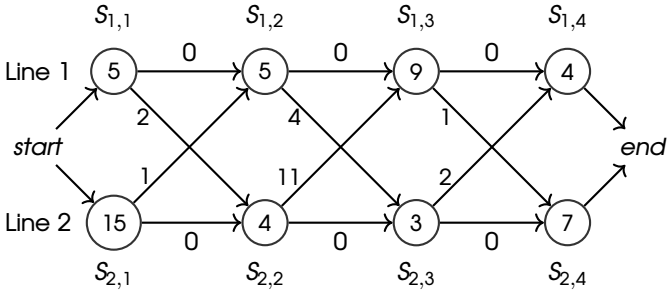
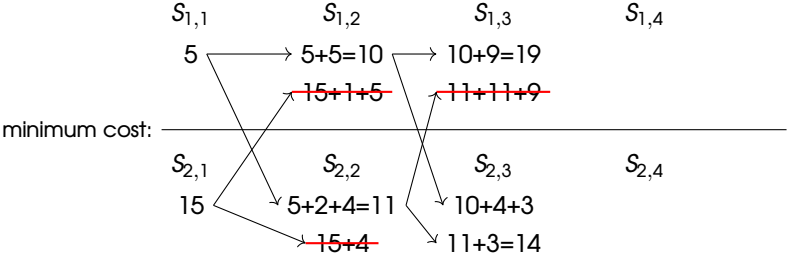
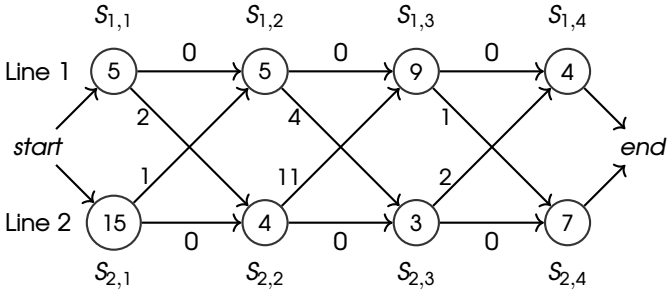
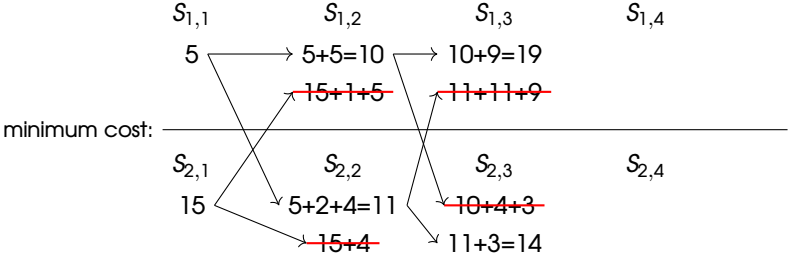


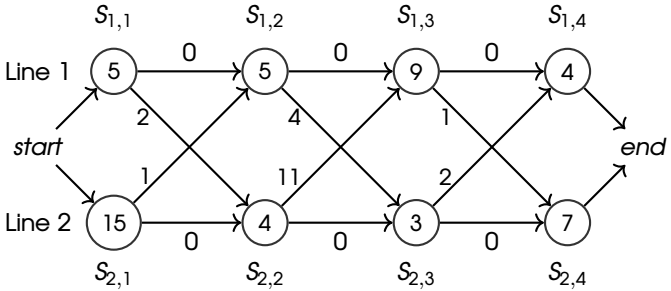


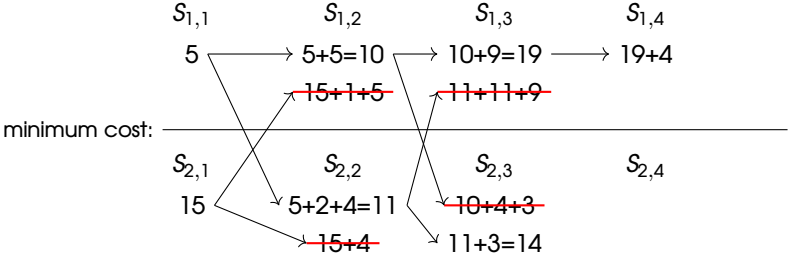
Example again



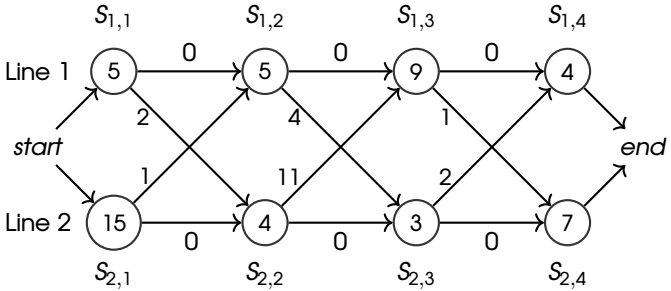


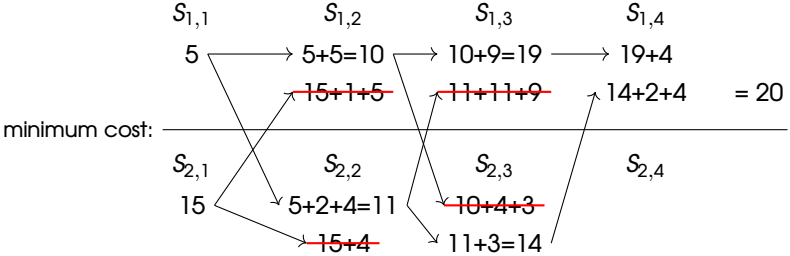
Example again



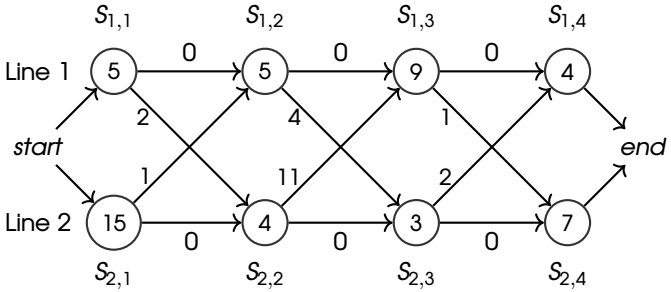


Example again





Example again

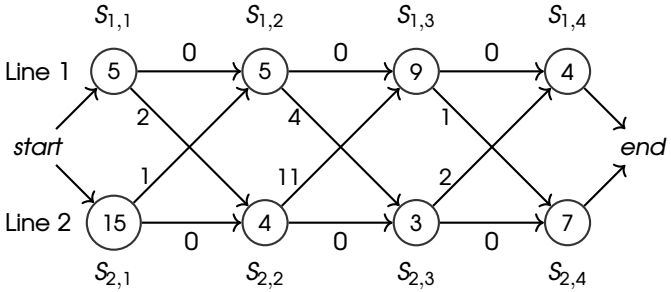


$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$	
5	5+5=10	10+9=19	~~19+4~~	
	~~15+1+5~~	~~11+11+9~~	14+2+4	= 20

minimum cost:

$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$
15	5+2+4=11	~~10+4+3~~	
	~~15+4~~	11+3=14	

Example again

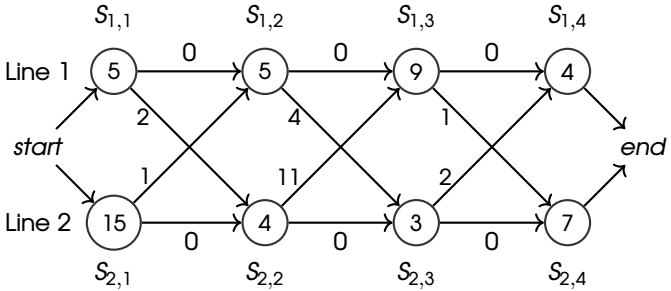


$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$	
5	5+5=10	10+9=19	~~19+4~~	
	~~15+1+5~~	~~11+11+9~~	14+2+4	= 20

minimum cost:

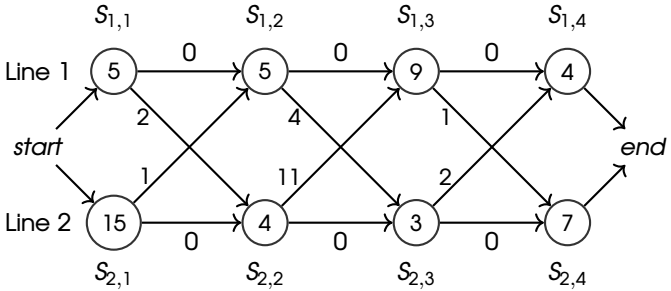
$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$
15	5+2+4=11	~~10+4+3~~	19+1+7
	~~15+4~~	11+3=14	

Example again



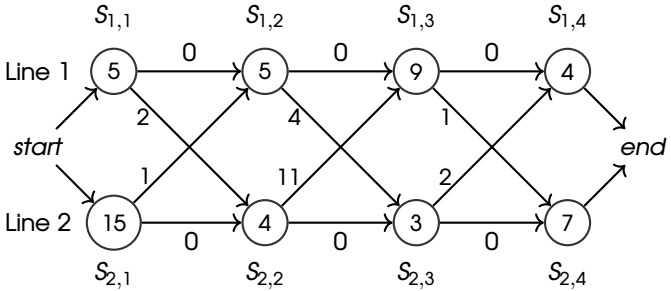
$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$	
5	5+5=10	10+9=19	~~19+4~~	
	~~15+1+5~~	~~11+11+9~~	14+2+4	= 20
minimum cost:				
$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$	
15	5+2+4=11	~~10+4+3~~	19+1+7	
	~~15+4~~	11+3=14	14+7	

Example again



	$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$	
	5	5+5=10	10+9=19	~~19+4~~	
		~~15+1+5~~	~~11+11+9~~	14+2+4	= 20
minimum cost:					
	$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$	
	15	5+2+4=11	~~10+4+3~~	~~19+1+7~~	
		~~15+4~~	11+3=14	14+7	= 21

Example again

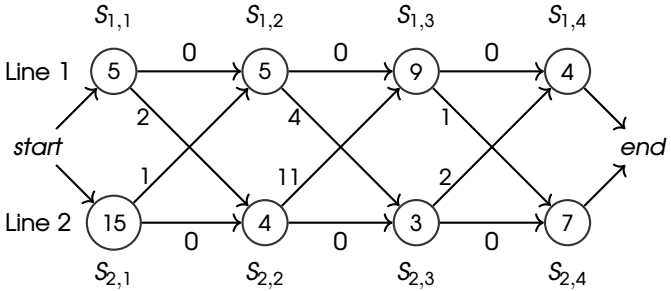


$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$	
5	5+5=10	10+9=19	~~19+4~~	
	~~15+1+5~~	~~11+11+9~~	14+2+4	= 20

minimum cost:

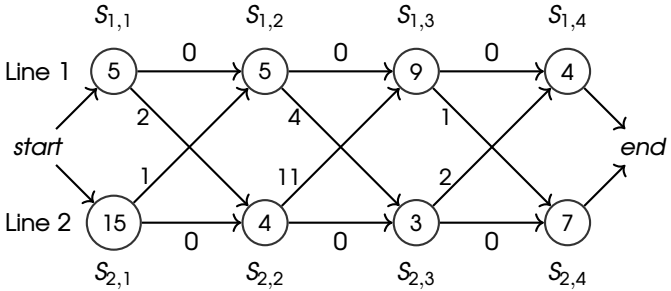
$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$	
15	5+2+4=11	~~10+4+3~~	~~19+1+7~~	
	~~15+4~~	11+3=14	14+7	~~= 21~~

Example again



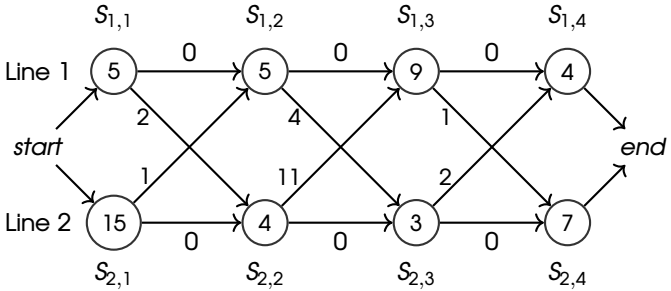
	$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$	
	5	5+5=10	10+9=19	~~19+4~~	
		~~15+1+5~~	~~11+11+9~~	14+2+4	= 20
minimum cost:					
	$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$	
	15	5+2+4=11	~~10+4+3~~	~~19+1+7~~	
		~~15+4~~	11+3=14	14+7	~~= 21~~

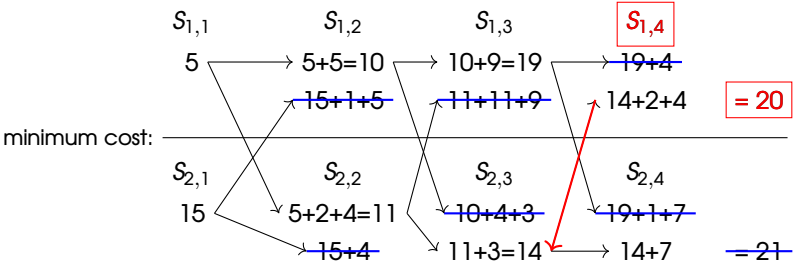
Example again



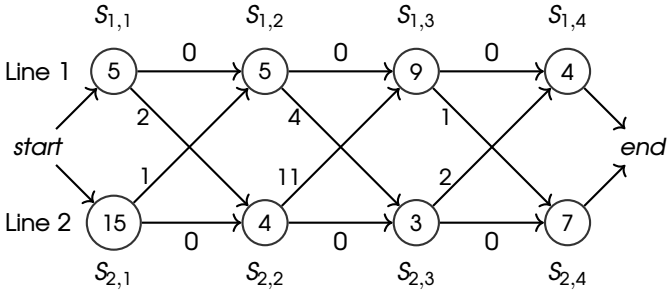
$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$	
5	5+5=10	10+9=19	~~19+4~~	
	~~15+1+5~~	~~11+11+9~~	14+2+4	$\boxed{= 20}$
minimum cost:				
$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$	
15	5+2+4=11	~~10+4+3~~	~~19+1+7~~	
	~~15+4~~	11+3=14	14+7	~~= 21~~

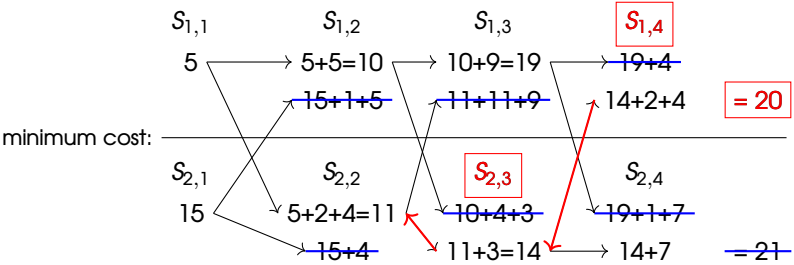
Example again



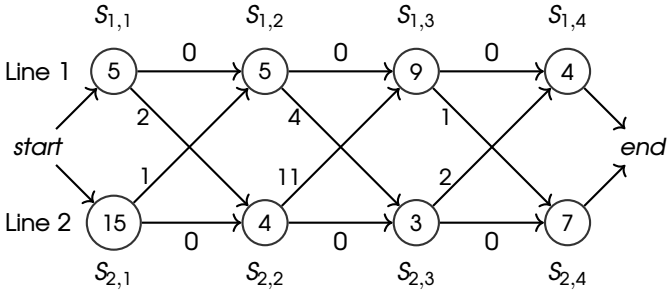


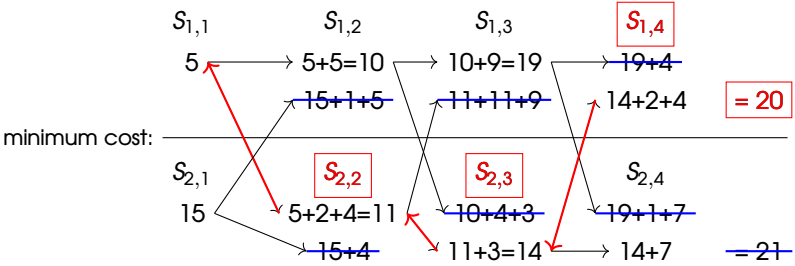
Example again



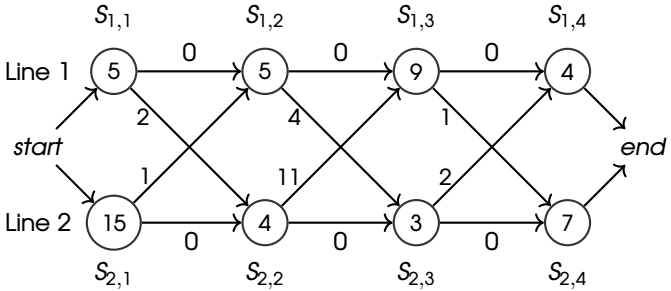


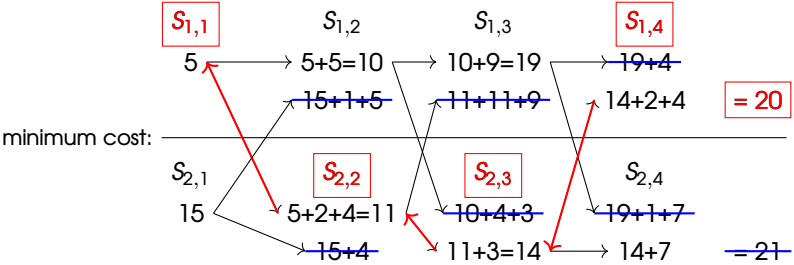
Example again



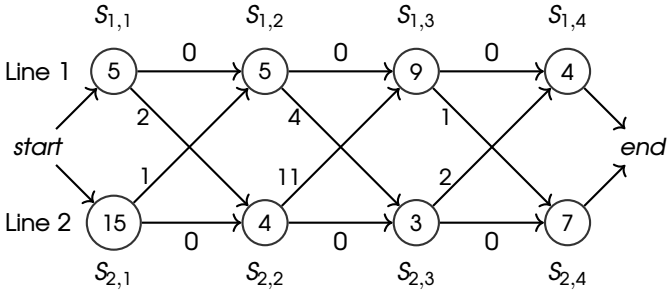


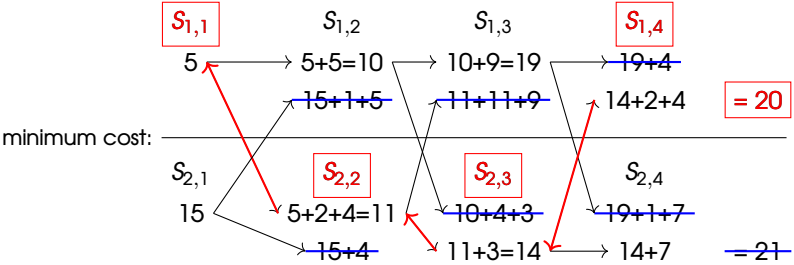
Example again



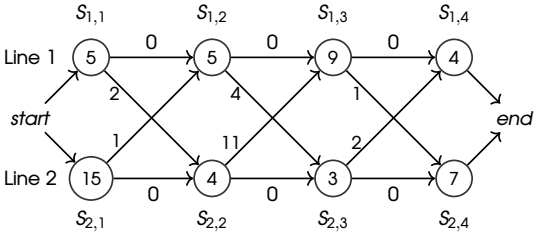


Example again - $s_{1,1}$, $s_{2,2}$, $s_{2,3}$, $s_{1,4}$





Summary on the Example



j	$f_1[j]$	$f_2[j]$
1	5	15
2		
3		
4		

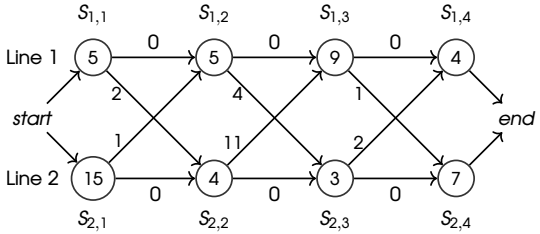
	$S_{1,1}$	$S_{1,2}$	$S_{1,3}$	$S_{1,4}$
	5			
minimum cost:				
	$S_{2,1}$	$S_{2,2}$	$S_{2,3}$	$S_{2,4}$
	15			

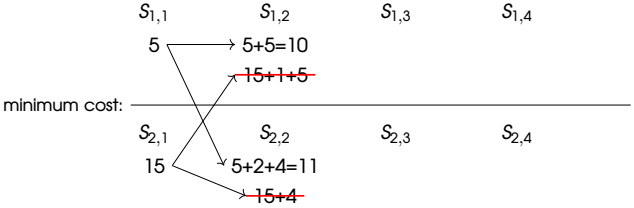
$$f_1[j] = \begin{cases} a_{1,1} & \text{if } j = 1 \\ \min\{f_1[j-1] + a_{1,j}, f_2[j-1] + t_{2,j-1} + a_{1,j}\} & \text{if } j > 1 \end{cases}$$

$$f_2[j] = \begin{cases} a_{2,1} & \text{if } j = 1 \\ \min\{\min\{f_1[j-1] + t_{1,j-1} + a_{2,j}, f_2[j-1] + a_{2,j}\} & \text{if } j > 1 \end{cases}$$

$$f^* = \min\{f_1[n], f_2[n]\}$$

Summary on the Example





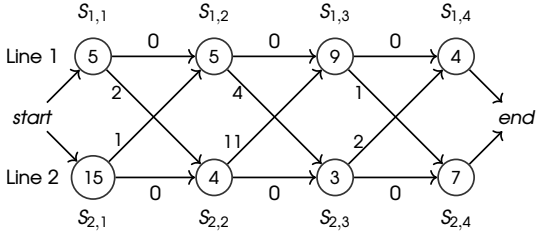
j	$f_1[j]$	$f_2[j]$
1	5	15
2	10	11
3		
4		

$$f_1[j] = \begin{cases} a_{1,1} & \text{if } j = 1 \\ \min\{f_1[j-1] + a_{1,j}, f_2[j-1] + t_{2,j-1} + a_{1,j}\} & \text{if } j > 1 \end{cases}$$

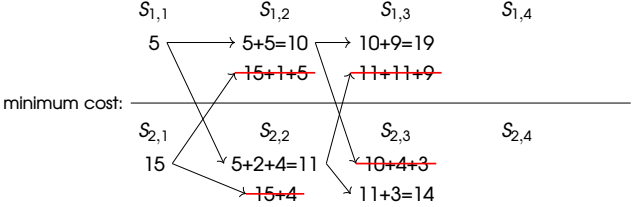
$$f_2[j] = \begin{cases} a_{2,1} & \text{if } j = 1 \\ \min\{\min\{f_1[j-1] + t_{1,j-1} + a_{2,j}, f_2[j-1] + a_{2,j}\} & \text{if } j > 1 \end{cases}$$

$$f^* = \min\{f_1[n], f_2[n]\}$$

Summary on the Example



j	$f_1[j]$	$f_2[j]$
1	5	15
2	10	11
3	19	14
4		

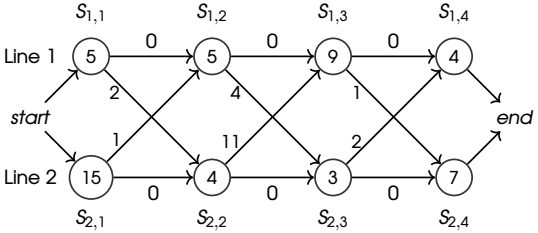


$$f_1[j] = \begin{cases} a_{1,1} & \text{if } j = 1 \\ \min\{f_1[j-1] + a_{1,j}, f_2[j-1] + t_{2,j-1} + a_{1,j}\} & \text{if } j > 1 \end{cases}$$

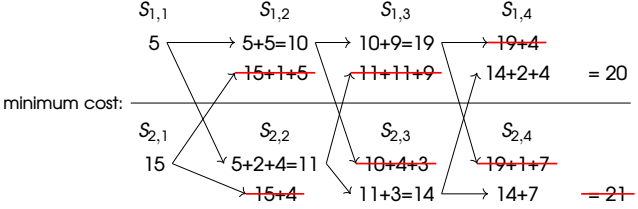
$$f_2[j] = \begin{cases} a_{2,1} & \text{if } j = 1 \\ \min\{\min\{f_1[j-1] + t_{1,j-1} + a_{2,j}, f_2[j-1] + a_{2,j}\} & \text{if } j > 1 \end{cases}$$

$$f^* = \min\{f_1[n], f_2[n]\}$$

Summary on the Example



j	$f_1[j]$	$f_2[j]$
1	5	15
2	10	11
3	19	14
4	20	21

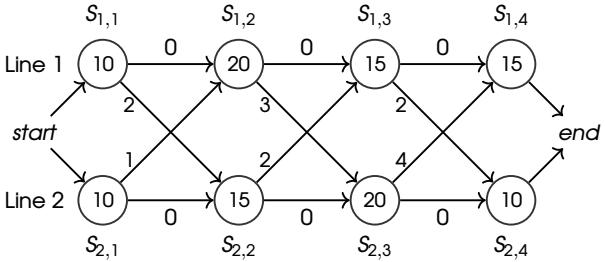


$$f_1[j] = \begin{cases} a_{1,1} & \text{if } j = 1 \\ \min\{f_1[j-1] + a_{1,j}, f_2[j-1] + t_{2,j-1} + a_{1,j}\} & \text{if } j > 1 \end{cases}$$

$$f_2[j] = \begin{cases} a_{2,1} & \text{if } j = 1 \\ \min\{\min\{f_1[j-1] + t_{1,j-1} + a_{2,j}, f_2[j-1] + a_{2,j}\} & \text{if } j > 1 \end{cases}$$

$$f^* = \min\{f_1[n], f_2[n]\}$$

Exercise



j	$f_1[j]$	$f_2[j]$
1		
2		
3		
4		

Pseudo code

$f_1[1] \leftarrow a_{1,1}$

$f_2[1] \leftarrow a_{2,1}$

for $j \leftarrow 2$ to n do

begin

$\quad f_1[j] \leftarrow \min\{f_1[j-1] + a_{1,j}, f_2[j-1] + t_{2,j-1} + a_{1,j}\}$

$\quad f_2[j] \leftarrow \min\{f_2[j-1] + a_{2,j}, f_1[j-1] + t_{1,j-1} + a_{2,j}\}$

end

$f^* \leftarrow \min\{f_1[n], f_2[n]\}$

What about 3 assembly lines?

Summary

Summary: Dynamic Programming for Assembly Line Scheduling

Next: Revision Lecture

For note taking